ARTHUR ACKER

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Data scientist with a strong foundation in statistics, machine learning, and data engineering. Experienced in building production-grade ETL pipelines, optimizing database performance, and shipping interactive analytical tools. Proven ability to translate complex data into actionable insights through clear storytelling and visualization, with applied interests in sports analytics and operational efficiency.

EDUCATION

University of Chicago — *M.S. in Applied Data Science* 2025 – 2026
GPA: 4.0/4.0 | Scholarship Recipient | Coursework: Machine Learning, Bayesian ML, Time Series, Big Data & Cloud Computing, Statistical Models

McGill University — *B.Sc. in Statistics & Computer Science* 2021 – 2025
GPA: 3.74/4.0 | James McGill Scholarship | Coursework: Applied ML, GLMs, Nonparametric Statistics, Algorithm Design, Database Systems

PROFESSIONAL EXPERIENCE

Capstone Researcher, Portfolio Optimization | *Pyxidr — UChicago Capstone* Mar 2026 – Aug 2026

- Backtested fixed-calendar vs. volatility-triggered reoptimization strategies in Python across a 499-trading-day window, finding a cadence with a 27.9% ($7.25M) gain over static buy-and-hold and a volatility signal matching it within 0.5% on far fewer trades.
- Built the BigQuery and Python ETL pipeline for a 303-bond, investment-grade universe sourced from Bloomberg to power the daily spread and duration data behind the optimizer.
- Designed and built a React/TypeScript dashboard backed by a Python FastAPI service, giving portfolio managers a live, optimizer-connected view of portfolio KPIs, suggested trades, risk, and constraint tracking in place of static reports.

Data Consultant | *Hutchinson Aerospace — Trenton, NJ* Sep 2025 – Aug 2026

- Retained following the internship below to provide ongoing maintenance and bug fixes for the database interface built during it, sustaining reliability for its 20+ engineer user base.

Data Science Intern | *Hutchinson Aerospace — Trenton, NJ* Sep 2024 – Aug 2025

- Redesigned inefficient SQL queries and introduced targeted indexing strategies, cutting search latency from ~90s to under 10s (approximately 90% improvement).
- Rebuilt a legacy database interface in Python for managing 10,000+ parts, inventory records, and engineering drawings, improving daily usability for a team of 20+ engineers.
- Developed automated data integrity scripts and debugging tools that reduced manual troubleshooting downtime by several hours per maintenance cycle.

Data Analytics Intern | *Hutchinson Aerospace — Burbank, CA* May – Aug 2024

- Built Python ETL pipelines to integrate and reconcile ERP and QMS data across departments, increasing cross-team data accuracy by 35%.
- Designed and implemented a centralized analytics layer to standardize fragmented multi-source data for scalable, self-serve reporting.
- Deployed a multi-tab interactive dashboard with dynamic visualizations that automated 5 recurring weekly reports for the 10-person Project Engineering team, replacing manual reporting.

ANALYTICS PORTFOLIO

Sports Analytics Dashboard (1st Place, McGill Hackathon) — Built an end-to-end data pipeline and R Shiny dashboard to visualize soccer player performance metrics from web-scraped data, presented to judges for first place.

Endurance Analytics Dashboard (Strava API) — Developed a Streamlit dashboard integrating Strava API data to analyze training load, fatigue, and race predictions using time-series analytics and rule-based models, visualized with Plotly.

Safest Route Directions (Concordia Hackathon) — Co-built a web application using Dash and graph algorithms to compute safety-optimized routes in Montreal, overlaying geospatial hex grids representing neighborhood safety scores.

Twitter AI Bot Detector — Applied NLP, cosine-similarity clustering, and feature engineering to identify bots in a large tweet dataset, achieving 90% accuracy.

Statistical Data Science Practicum — Led applied research involving sampling design, non-parametric statistics, and experimental analysis using R and Python. Presented findings to partner company leadership.

TECHNICAL SKILLS

Programming: Python, R, SQL, Bash, Java
ML & Data: scikit-learn, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, Streamlit
BI & Visualization: Tableau, Power BI, R Shiny, Plotly
Databases & Tools: PostgreSQL, MySQL, Git, Docker, Jupyter
Core Competencies: Machine Learning, Predictive Modeling, Statistical Inference, Data Engineering, Experimental Design, NLP
Languages: French (Native), Spanish (Intermediate)